

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(g)		Charge of $\bar{u}d = -\frac{2}{3} - \frac{1}{3}$	1			1		
	(h)		Either: Mass = $\frac{172 \text{ G[eV]}}{931 \text{ M[eV u}^{-1}]} = 185 \text{ [u]} (1)$ $\therefore \text{Mass} = 185 \text{ [u]} \times 1.66 \times 10^{-27} \text{ [kg u}^{-1}] = 3.07 \times 10^{-25} \text{ [kg]} (1)$ Or: Mass energy = $172 \text{ GeV} \times 1.60 \times 10^{-19} \text{ J eV}^{-1} = 2.75 \times 10^{-8} \text{ [J]} (1)$ $\therefore \text{Mass} = \frac{2.75 \times 10^{-8} \text{ [J]}}{(3.00 \times 10^8 \text{ [m s}^{-1}])^2} = 3.06 \times 10^{-25} \text{ [kg]} (1)$		2		2	2	
			Question 7 total	3	8	9	20	12	0